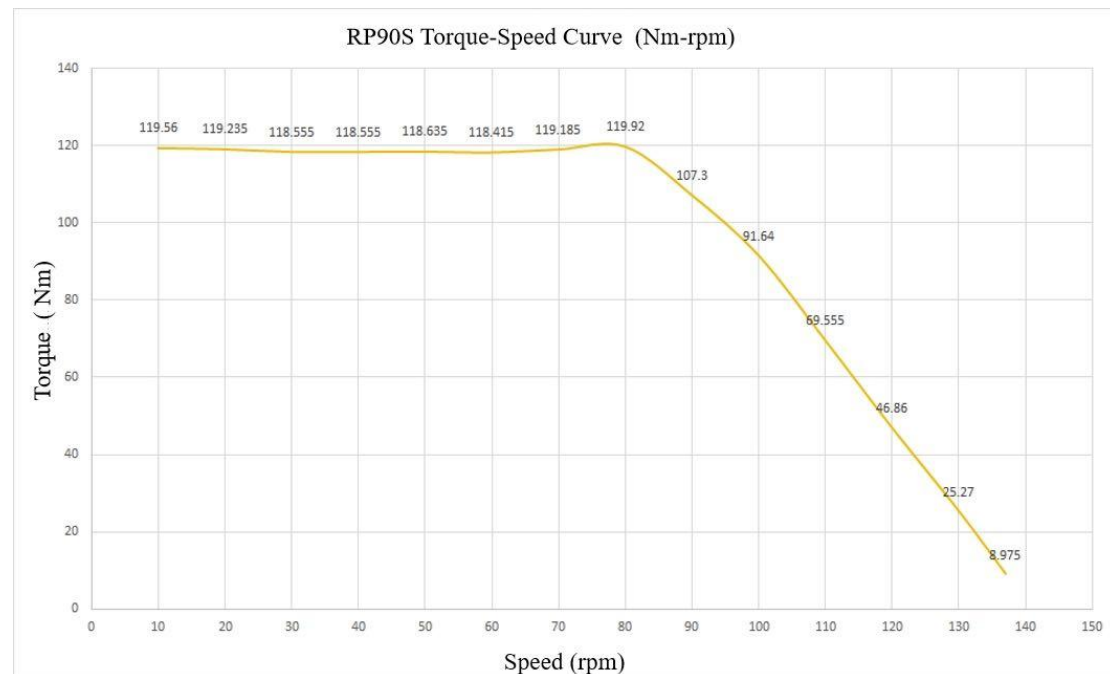


1. Torque-speed data

1.1 Peak torque-speed curve at 48 V



1.2 Continuous/rated torque-speed curve at 48 V

A: During testing, it can be provided later

1.3 Torque-speed curves for different duty cycles, if available

A: No relevant tests have been arranged

1.4 Test conditions of the provided torque-speed curve, including ambient temperature, housing temperature, cooling condition, and measurement setup

A: Ambient temperature: 26°C

Housing temperature: No monitored housing temperature data

Cooling condition: Natural air cooling (no forced cooling equipment applied)

Measurement setup: Joint fixed rigidly on test bench; load controlled by dynamometer, speed increases in 10% of max speed increments, forward & reverse rotation tested separately, record the maximum sustainable torque at each speed point to generate torque-speed curves.

2. Duty cycle and thermal limits

2.1 Maximum duration for peak torque operation, especially at 120 Nm

A: During testing, it can be provided later

2.2 Allowed duty cycle for 120 Nm, 60 Nm, and rated torque

A: No relevant tests have been arranged yet

2.3 Thermal derating curves

A: Not have

2.4 Winding and housing temperature limits

A: During testing, it can be provided later

2.5 Thermal time constants, if available

A: Not have

2.6 Location of the temperature sensor used by the driver

A: Two sensors: one beside the MOS, another embedded in motor winding.

3. Friction and backdrivability

3.1 Static friction / breakaway torque at the output

A: During testing, it can be provided later

3.2 Coulomb friction & 3.3 Viscous friction coefficient

During the Backdrive torque test, the torque fluctuation value and the average Backdrive torque can be output, which can be used as a reference.

3.4 Stribeck friction data near zero speed, if available

A: Not have

3.5 No-load current versus output speed

A: During testing, it can be provided later

3.6 Backdrive torque versus speed

A: During testing, it can be provided later

3.7 Friction data for both rotation directions and different temperatures, if available

A: Not have

4. Gearbox stiffness, backlash, and compliance

4.1 Torsional stiffness of the gearbox/output stage

A: Pending confirmation, awaiting feedback from the gearbox manufacturer.

4.2 Torque-deflection curve

A: Not have

4.2 Gear damping

A: To be confirmed

4.4 Backlash hysteresis or lost-motion curve

A: During testing, it can be provided later

4.5 Measurement method for the specified backlash value

A: The backlash test is implemented in accordance with Clause 5.4.6 of GB/T 43200-2023, with detailed steps as below:

1. Lock the joint under braking status;
2. Apply load on the output side: start from zero torque, gradually load in one direction up to rated torque T_N , then reversely load slowly to $-T_N$, reload forward to T_N again, followed by gradual unloading. Record output angular displacement corresponding to each applied torque throughout the full cycle to plot the hysteresis curve;
3. Calculate backlash from the hysteresis curve: extract median angular values at $+3\%T_N$ and $-3\%T_N$ respectively, and the difference between the two median values is defined as joint backlash.

4.6 Output bearing stiffness, if available

A: To be confirmed

5. Motor and current model

5.1 Motor torque constant K_t

A: $K_t=0.16\text{Nm/A}$

5.2 Output torque per phase current or command current

A: 2.4NM/A

5.3 Maximum phase current

A: 49Arms

5.4 RMS phase current limit

A: The RMS phase current limit is defined based on peak current value.

For RP90L model, the corresponding peak current is 49Arms .

5.5 Bus current limit

A: 12.5Arms

5.6 Current saturation behavior

A: Upon current saturation, the system activates current limiting protection

5.7 Torque command scaling and units

A: Unit: thousandths (‰) of the rated torque

6. Driver and control-loop dynamics

6.1 Current/torque control loop frequency

A: 16k HZ

6.2 Current/torque control bandwidth

A: 1.2KHZ

6.3 Position and velocity loop frequency

A: 8KHZ

6.4 Command-to-torque latency

A: Millisecond level

6.5 Internal filtering of torque, velocity, or position commands

A: Not open to the public

6.6 Torque step response

A: Not open to the public

6.7 Recommended maximum command update rate over CAN FD

A: Depending on the number of stations, it is recommended to use 4ms for seven axes

6.8 Exact implementation of MIT control mode, including formula, units, and limits

A: RP MIT mode parameter ranges:

1.Target position: 0~65535 $\rightarrow$ -12.5 rad ~ 12.5 rad

2.Target velocity: 0~4095 $\rightarrow$ -60 rad/s ~ 60 rad/s

3.Kp: 0~4095 $\rightarrow$ 0.0 ~ 500.0

4.Kd: 0~4095 $\rightarrow$ 0.0 ~ 5.0

5.Target torque: 0~4095 $\rightarrow$ -32 Nm ~ 32 Nm

205	0x2140	0x0	Mit P Min	3242721280	3242721280	
206	0x2141	0x0	Mit P Max	1095237632	1095237632	
207	0x2142	0x0	Mit V Min	3262119936	3262119936	
208	0x2143	0x0	Mit V Max	1114636288	1114636288	
209	0x2144	0x0	Mit KP Min	0	0	
210	0x2145	0x0	Mit KP Max	1140457472	1140457472	
211	0x2146	0x0	Mit KD Min	0	0	
212	0x2147	0x0	Mit KD Max	1084227584	1084227584	
213	0x2148	0x0	Mit I Min	3254779904	3254779904	
214	0x2149	0x0	Mit I Max	1107296256	1107296256	

These object dictionaries can set the threshold range for the mit parameter, as shown in the above figure

MIT模式控制参数(MIT Control Parameters) 0x2130 (byte0-byte3) 0x2131 (byte4-byte7) 小端模式 (Little Endian)																				
		bit (位)								颜色 Color	变量 Variables									
byte (字节)	byte0	bit7		bit6		bit5		bit4		bit3		bit2		bit1		bit0			MIT Target Position (Big-Endian)	
	byte1	bit15		bit14		bit13		bit12		bit11		bit10		bit9		bit8			MIT Target Speed (Big-Endian)	
	byte2	bit23		bit22		bit21		bit20		bit19		bit18		bit17		bit16			MIT Kp (Big-Endian)	
	byte3	bit31		bit30		bit29		bit28		bit27		bit26		bit25		bit24			MIT Kd (Big-Endian)	
	byte4	bit39		bit38		bit37		bit36		bit35		bit34		bit33		bit32			MIT Target Torque (Big-Endian)	
	byte5	bit47		bit46		bit45		bit44		bit43		bit42		bit41		bit40				
	byte6	bit55		bit54		bit53		bit52		bit51		bit50		bit49		bit48				
	byte7	bit63		bit62		bit61		bit60		bit59		bit58		bit57		bit56				

7.4 Velocity estimation method

A: RP Series CanopenTool Host Conversion Equations

$$\text{Target Speed (rpm)} = \text{Pulse Value} \div 524288 \times 60$$

$$\text{Acceleration (rpm/s)} = \text{Pulse Value} \div 524288 \times 60$$

$$\text{Target Angular Position (}^\circ\text{)} = \text{Pulse Value} \div 524288 \times 360^\circ$$

$$\text{Target Output Torque (N}\cdot\text{m)} = \text{Cmd_Torque} \times \text{T_rated} / 1000$$

7.5 Position and velocity update rate

A: 8kHz

7.6 Encoder and measurement latency

A: There is no specific value.

7.7 Relationship between input-side and output-side encoder measurements

A: The input-side encoder is 17-bit mounted on motor rotor, and the output-side flange encoder is 19-bit.

8. Efficiency and regeneration

8.1 Efficiency map versus torque, speed, and direction

A: During testing, it can be provided later

8.2 Motoring and braking/regeneration behavior

A: No energy regeneration

8.3 Regenerative current limits

A: No energy regeneration

8.4 Bus overvoltage behavior during braking

A: Overvoltage protection

8.5 Requirements for brake resistor or external protection, if applicable

A: 1000W discharge module

9. Mechanical limits for legged robot use

9.1 Maximum impact torque

A: 120NM

9.2 Maximum external bending moment at the output

A: Not have

9.3 Shock load limits

A: Radial (dynamic/static) allowable load: 2810/4680N

Axial (dynamic/static) allowable load: 6440/30480N

9.4 Radial and axial load limits versus lifetime

A: Not have

9.5 Recommended mounting stiffness

A: The joint mounting interface is manufactured from aluminum alloy.

9.6 Lifetime data under cyclic legged-robot-type loads, if available

A: Not have

10. Communication and fault behavior

10.1 Full CANopen / CAN FD protocol manual & 10.2 Object dictionary



EYou-RP系列关
节CANopen通信

A: RP Series communication manual

10.3 MIT mode packet format

A: Please refer to section 3.8 of the communication manual for specific information

205	0x2140	0x0	Mit P Min	3242721280	3242721280	
206	0x2141	0x0	Mit P Max	1095237632	1095237632	
207	0x2142	0x0	Mit V Min	3252119936	3252119936	
208	0x2143	0x0	Mit V Max	1114636288	1114636288	
209	0x2144	0x0	Mit KP Min	0	0	
210	0x2145	0x0	Mit KP Max	1140457472	1140457472	
211	0x2146	0x0	Mit KD Min	0	0	
212	0x2147	0x0	Mit KD Max	1084227584	1084227584	
213	0x2148	0x0	Mit T Min	3254779904	3254779904	
214	0x2149	0x0	Mit T Max	1107296256	1107296256	

10.4 Command units and scaling

A: Each dictionary is different, please refer to the manual for details.

10.5 Watchdog behavior

A: Currently, there is no watchdog function set on the software

10.6 Error codes and fault handling

A: Please refer to the manual for details.

10.7 Behavior when torque, speed, current, voltage, or temperature limits are exceeded

Enable when any parameter of torque, speed, current, bus voltage or winding temperature exceeds the limit and an error is reported.

11. Mass properties / CAD data

11.1 Detailed CAD model



RP90S外形图
V1.0 20260310.S

A: The STEP of RP90S

11.2 Exact mass properties

A: During testing, it can be provided later

11.3 Center of gravity

A: Symmetrical structure

11.4 Rotor inertia

A: $9.33 \times 10^{-5} \text{ kg} \cdot \text{mm}^2$

11.5 Output inertia

A: $269.40 \text{ kg} \cdot \text{mm}^2$

11.6 Mounting interface stiffness, if available

A: Mounting interface is made of aluminum alloy.